T Pranav Kutralingam

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Technical Skills

Languages: Python, Java, C++, C#, C, SQL, TypeScript, JavaScript, Rust
AI & ML: PyTorch, scikit-learn, NumPy, Pandas, Reinforcement Learning, Model Evaluation, Benchmarking
LLMs & GenAI: OpenAI Agent SDK, OpenAI API, Hugging Face, Retrieval-Augmented Generation (RAG), pgvector, LLM-as-a-Judge Evaluation, Prompt Engineering, Fine-Tuning
Backend & Web: FastAPI, Flask, Spring Boot, Microservices, REST APIs, React
Infrastructure & DevOps: Docker, Kubernetes, Helm, Jenkins, GitHub Actions, Continuous Integration/Continuous Deployment (CI/CD), Slurm, AWS, Azure, PostgreSQL, Splunk, Grafana, Locust, Linux

Experience

Product Engineer Intern, Pretorin Inc. (AI Compliance Automation Platform) *Mar 2026 – Present*

- Owned a Python data-freshness pipeline keeping 26 compliance frameworks (like NIST, FedRAMP, SOC 2) synced to upstream sources for a custom GPT-5.5 based hybrid RAG architecture.
- Designed an LLM evaluation gate in CI: deterministic schema checks plus a calibrated LLM-as-judge benchmarked against a hand-labeled golden set, failing closed to block bad AI enrichments before release.
- Built a production attestation service that cryptographically signs AI- and human-produced compliance evidence, making every record independently verifiable and hash-bound to its source.
- Built the platform's first Kubernetes load/stress framework (Locust, Helm, 5 scenarios, 100 concurrent users) with Grafana dashboards and an OpenAI token-budget kill-switch, catching latency and AI-cost regressions before release.

Graduate Research Assistant, DaRL Labs (ASU) *May 2024 – Dec 2025*

- Designed fault-tolerant batch execution pipelines in Python (checkpointing, retries, idempotency) that ran 2,000+ PyTorch experiments across a 10-GPU Slurm cluster with zero manual restarts.
- Cut experiment turnaround from 4-5 days to overnight by designing a batch execution pipeline on Slurm that automated result collection and inference.
- Built the Python data and integration layer for MLCL, a multilingual LLM tool-calling benchmark, turning manual prep and ad hoc runs into one-command batched evaluation with automated scoring.

Software Engineer, Amadeus Labs *Aug 2021 – Dec 2023*

- Redesigned a distributed E2E test execution pipeline on Kubernetes and Jenkins with parallel execution and focused jobs, cutting total runtime from 12 hours to 4 hours.
- Cut manual triage effort by 60% across 15K+ events/month by building an NLP-based error-log monitoring system in Python that tagged genuine failures and routed them to owning teams.
- Designed observability pipelines and Splunk dashboards to detect anomalies and surface production issues and alert the relevant code owners before customer impact.
- Owned backend production services in Java and Python powering high-volume, customer-facing travel workflows with strict uptime, latency, and reliability requirements.

Software Engineer Intern, Amadeus Labs *Jan 2021 – May 2021*

- Modernized a legacy monolithic JSP application into a Spring Boot microservices platform, improving fault isolation, deployment safety, and scalability.
- Cut backend scheduler and parser runtime by 30% through async execution and targeted refactors in Java.
- Improved API response times by 45% under peak production load through SQL query tuning and caching.

Projects and Publications

Latent Adaptation of Foundation Policies for Sim-to-Real Transfer *Published, ICLR 2026*

- Adapts pretrained foundation policies to real-world dynamics at deployment via latent-space adaptation, without retraining.

Lost in Execution: On the Multilingual Robustness of Tool Calling in LLMs *Published, ACL 2026*

- Introduced MLCL, a multilingual tool-calling benchmark; found parameter-value language mismatch is the dominant execution failure mode.

HEAL AI (React, FastAPI, PostgreSQL, RAG, Docker) *Hackathon 2025*

- Parses insurance documents and flags medical-bill overcharges using a RAG pipeline over vector embeddings.

Scalable Cloud-Based Video & Image Analysis Platform (AWS, Docker, PyTorch) *2025*

- Cloud-native ML inference pipeline on AWS reaching sub-2s latency at 80%+ lower infrastructure cost.

Education

Arizona State University – MS in Computer Science, GPA 3.97/4.0 *Jan 2024 – Dec 2025*
SRM Institute of Science and Technology – BTech in Computer Science, GPA 3.7/4.0 *May 2017 – May 2021*